# Eliminating Child Care Deserts in New York State through Optimization

## The Problem of Budgeting

### Objective:

Minimize total funding so that no ZIP is a child care desert, and age 0–5 coverage meets policy.

Total Funding = New construction cost + expansion cost + 0–5% special equipment fee

## 1. Sets and Indices

Index

$z$ : Index for each ZIP code in New York State

$f$ : Index for each existing child care facility

$s$ : Index for new facility size: $s \in \{\text{Small}, \text{Medium}, \text{Large}\}$.

## 2. Parameters

- $S_{0-12z}$: Existing number of available slots for children aged 0-5 (infant + toddler + preschool) in Zip $z$
- $S_{0-12z}$: Existing number of total regulated slots (children aged 0-12) in Zip $z$
- $P_{0-5z}$: number of children aged 0-5
- $P_{0-12z}$: number of children aged 0-12
- $P_{5-12z}$: number of children aged 5-12
- $I_z$: average income for every ZIP $z$ (data from: avg_individual_income.csv)
- $e_z$: employment rate in ZIP $z$ (data from: employment_rate.csv)
- $n_f$: facility $f$'s current capacity(slots)
  -Build new one
- $C_s$ : Fixed cost to build a new facility of size $s$
- $\text{Cap}_s$: Total capacity (slots) of a new facility of size $s$
- $\text{Cap}_{0-5,s}$: Max capacity for ages 0–5 slots in a new facility of size $s$

| Facility Size | Max # of Slots (Ages 0-5) | Cost of New Facility ($) to NYS |
|---|---|---|
| 100 slots (Small) | 50 slots | 65,000 |
| 200 slots (Medium) | 100 slots | 95,000 |
| 400 slots (Large) | 200 slots | 115,000 |

-Expansion

- $C_{\text{expand, fixed}}$ = 20,000: Baseline cost for facility expansion
- $C_{\text{expand, unit}}$=200: Capacity-based cost per existing slot for expansion
- $C_{0-5}$=100: Special equipment fee per new slot (expansion)
- $L_{\text{cap}}$ : Maximum expansion limit per facility (min 500/120% **$n_f$)

# 3. Decision Variables

- $x_{z,s}$: Number of new facilities of size $s$ to build in ZIP $z$
- $x_f$: Total number of slots added via expansion at facility $f$
- $x_{f0-5}$: Number of $0-5$ slots added via expansion at facility $f$
- $Y_z$=1, if Zip $z$ is a high-demand area, $0$ otherwise(High-Demand Area Definition: either of the following conditions is met: The employment rate $\geq 60\%$**or**The average individual income $\leq \$60,000$)
- $\delta_f$=1 if facility $f$ is expanded ($x_f > 0$), $0$ otherwise
- $S_{newz}$: Total new slots added in ZIP $z$ (from expansion and new construction)
- $S_{new,0-5,z}$: Total new $0-5$ slots added in ZIP $z$

# 4. Objective Function

### Minimize Total Funding

The objective function minimizes the sum of new construction costs, fixed/capacity-based expansion costs, and the special equipment fees for new $0-5$ slots.

$$\min Z = \sum_z \left[ \sum_s (C_s \cdot x_{z,s}) + \sum_{f \in z} ((C_{\text{expand, fixed}} + C_{\text{expand, unit}} n_f) \cdot \delta_f + C_{0-5} \cdot x_{f0-5}) \right]$$

# 5. Constraints

## A. Slot Capacity Linkage

1. Total New Slots in ZIP $z$: Relates the total new slots $S_{newz}$ to the decision variables $x_f$ and $x_{z,s}$.

$$\forall z: \quad S_{newz} = \sum_{f \in z} x_f + \sum_s (\text{Cap}_s \cdot x_{z,s})$$

2. Total New $0-5$ Slots in ZIP $z$: Relates new $0-5$ slots $S_{new,0-5,z}$ to $x_{f0-5}$ and $\text{Cap}_{0-5,s}$.

$$\forall z: \quad S_{new,0-5,z} = \sum_{f \in z} x_{f0-5} + \sum_s (\text{Cap}_{0-5,s} \cdot x_{z,s})$$

## B. Demand Classification and Coverage

3. High-Demand Area Determination ($Y_z$):

   $Y_z = 1$ if ($e_z \geq 0.60$ OR $I_z \leq \$60,000$), $0$ otherwise.
4. **Non-Desert Condition (0–12 Coverage):** Ensures that the total available slots (existing + new) meets the minimum requirement based on the demand classification of ZIP $z$.
   - **High-Demand Target:** $1/2 \times P_{0-12z}$
   - **Normal-Demand Target:** $1/3 \times P_{0-12z}$

$$\forall z: \quad S_{0-12z} + S_{newz} \geq Y_z \cdot \frac{1}{2} P_{0-12z} + (1 - Y_z) \cdot \frac{1}{3} P_{0-12z}$$

5. NYS Policy Requirement (0–5 Coverage): Ensures that capacity for young children meets the state policy target.

$$\forall z: \quad S_{0-5z} + S_{new,0-5,z} \geq \frac{2}{3} P_{0-5z}$$

## C. Facility Expansion Limits and Costs

6. Expansion Quantity Limit: Limits the expanded slots $x_f$ to a maximum of 500 or 120% of current capacity, whichever is smaller.

$$\forall f: \quad 0 \leq x_f \leq \min(L_{\text{cap}}, 1.2 \cdot n_f)$$

7. 0-5 Slots within Expansion: Ensures that the slots dedicated to the $0-5$ age group are a subset of the total expanded slots.

$$\forall f: \quad 0 \leq x_{f0-5} \leq x_f$$

8. Expansion Cost Activation ($\delta_f$): Activates the fixed/capacity-based expansion cost when any expansion occurs.

$$\forall f: \quad x_f \leq \min(L_{\text{cap}}, 1.2 \cdot n_f) \cdot \delta_f$$

# The Problem of Realistic Capacity Expansion and Location

## 1. Sets and Indices

- $\mathcal{Z}$: set of ZIP code areas, indexed by $z$
- $\mathcal{F}$: set of existing child care facilities, indexed by $f$; each belongs to area $a(f) \in \mathcal{Z}$
- $\mathcal{L}_z$: set of candidate locations for new facilities in area $z$, indexed by $i$
- $\mathcal{S} = \{\text{Small}, \text{Medium}, \text{Large}\}$: set of possible new facility sizes, indexed by $s$
- $\mathcal{P}_z$: set of conflicting pairs $(i, j)$ of candidate sites in area $z$ such that their distance $d_{ij} < 0.06$ miles
- $\mathcal{Q}_z$: set of conflict pairs between candidate site $i \in \mathcal{L}_z$ and existing facility $f$ such that $d_{if} < 0.06$ miles

## 2. Parameters

### Population and Policy Thresholds

- $\mathrm{P}_z^{\text{all}}$: total number of children (age 0–12) in area $z$
- $\mathrm{P}_z^{0\text{–}5}$: number of children aged 0–5 in area $z$
- $\mathrm{HighDem}_z \in \{0, 1\}$: 1 if area $z$ is high-demand (employment $\geq 60\%$ or income $\leq \$60\text{k}$)
- $r_z^{\text{desert}} = 0.5 \cdot \mathrm{HighDem}_z + \frac{1}{3}(1 - \mathrm{HighDem}_z)$: desert elimination ratio threshold
- $r^{0\text{–}5} = \frac{2}{3}$: coverage requirement for children aged 0–5

### Existing Facilities

- $n_f$: current capacity (number of slots) of facility $f$
- $n_f^{0\text{–}5}$: current number of slots for children aged 0–5 at facility $f$
- $B_z^{0\text{–}12}$: Existing capacity in ZIP $z$ for ages 0–12 (sum of $n_f$ for $f$ in $z$).
- $B_z^{0\text{–}5}$: Existing capacity in ZIP $z$ for ages 0–5 (sum of $n_f$ for $f$ in $z$ - assuming total capacity can serve 0–5).
- $\overline{x}_f = 0.2 n_f$: maximum expansion allowed ($20\%$ of existing capacity)

### Expansion Cost Function (Piecewise Convex)

For each facility $f$, the cost $c_{\text{ext},f}(x_f)$ depends on the scale of expansion:

$$c_{\text{ext},f} = \begin{cases} (20,000 + 200n_f) \cdot \frac{x_f}{n_f}, & 0 < x_f/n_f \leq 0.1, \\ (20,000 + 400n_f) \cdot \frac{x_f}{n_f}, & 0.1 < x_f/n_f \leq 0.15, \\ (20,000 + 1000n_f) \cdot \frac{x_f}{n_f}, & 0.15 < x_f/n_f \leq 0.2. \end{cases}$$

Larger expansions are significantly more expensive, and expansions beyond **20%** are prohibited.

We linearize this convex function using three segments:

$$U_f^{(1)} = 0.1n_f, \quad U_f^{(2)} = 0.05n_f, \quad U_f^{(3)} = 0.05n_f$$

with corresponding unit costs:

$$c_f^{(1)} = \frac{20,000 + 200n_f}{n_f}, \quad c_f^{(2)} = \frac{20,000 + 400n_f}{n_f}, \quad c_f^{(3)} = \frac{20,000 + 1000n_f}{n_f}.$$

### New Facility Specifications

| Size | Total Capacity | 0–5 Slots Max | Construction Cost |
|---|---|---|---|
| Small | 100 | 50 | $65,000 |
| Medium | 200 | 100 | $95,000 |
| Large | 400 | 200 | $115,000 |

### Equipment Cost for 0–5 Slots

Each new slot for children aged **0–5** costs an additional **$100**.

## 3. Decision Variables

### Expansion Variables

- $x_f^{(k)} \geq 0$: number of slots expanded in segment $k$ for facility $f$
- $x_f = \sum_{k=1}^{3} x_f^{(k)}$: total expansion at facility $f$
- $x_f^{0\text{–}5} \geq 0$: number of expanded slots for $0$–$5$ children; $x_f^{0\text{–}5} \leq x_f$

### New Construction Variables

- $y_{izs} \in \{0, 1\}$: $1$ if a facility of size $s$ is built at candidate site $i$ in area $z$
- $b_{izs}^{0\text{–}5} \geq 0$: number of $0$–$5$ slots in that facility (must satisfy $b_{izs}^{0\text{–}5} \leq \text{Cap}_s^{0\text{–}5} y_{izs}$)

## 4. Objective Function

$$\min \underbrace{\sum_{z} \sum_{i \in \mathcal{L}_z} \sum_{s \in \mathcal{S}} \text{Cost}_s^{\text{build}} y_{izs}}_{\text{Construction Cost}} + \underbrace{\sum_{f} \sum_{k=1}^{3} c_f^{(k)} x_f^{(k)}}_{\text{Expansion Cost}} + \underbrace{100 \cdot \left( \sum_{f} x_f^{0\text{–}5} + \sum_{z,i,s} b_{izs}^{0\text{–}5} \right)}_{\text{Equipment Cost for 0–5 Slots}}.$$

The goal is to **minimize the total funding** needed to eliminate child care deserts while respecting expansion and distance constraints.

# 5. Constraints

## (A) Expansion Feasibility

$$\begin{aligned} 0 \le x_f^{(k)} \le U_f^{(k)}, \quad k = 1,2,3, \forall f \\ x_f = \sum_{k=1}^{3} x_f^{(k)} \le 0.2 n_f, \quad \forall f \\ 0 \le x_f^{0\text{–}5} \le x_f, \quad \forall f \end{aligned}$$

## (B) New Facility Feasibility

$$\begin{aligned} 0 \le b_{izs}^{0\text{–}5} \le \text{Cap}_s^{0\text{–}5} y_{izs}, \quad \forall z, i, s \\ \sum_{s \in \mathcal{S}} y_{izs} \le 1, \quad \forall z, i \end{aligned}$$

## (C) Capacity Accounting

For each area $z$:

$$\begin{aligned} \text{Add}_z^{\text{all}} &= \sum_{f : a(f) = z} x_f + \sum_{i \in \mathcal{L}_z} \sum_{s} \text{Cap}_s y_{izs}, \\ \text{Add}_z^{0\text{–}5} &= \sum_{f : a(f) = z} x_f^{0\text{–}5} + \sum_{i \in \mathcal{L}_z} \sum_{s} b_{izs}^{0\text{–}5}, \\ \text{Cap}_z^{\text{all}} &= \text{B}_z^{\text{all}} + \text{Add}_z^{\text{all}}, \\ \text{Cap}_z^{0\text{–}5} &= \text{B}_z^{0\text{–}5} + \text{Add}_z^{0\text{–}5}. \end{aligned}$$

## (D) Coverage and Desert Elimination

$$\text{Cap}_z^{\text{all}} \ge r_z^{\text{desert}} \text{P}_z^{\text{all}}, \quad \text{Cap}_z^{0\text{–}5} \ge r^{0\text{–}5} \text{P}_z^{0\text{–}5}, \quad \forall z$$

equalize to

$$\text{Base}_z^{\text{all}} + \sum_{f : a(f) = z} x_f + \sum_{i,s} \text{Cap}_s y_{izs} \ge r_z^{\text{desert}} \text{Pop}_z^{\text{all}},$$

$$\text{Base}_z^{0\text{–}5} + \sum_{f : a(f) = z} x_f^{0\text{–}5} + \sum_{i,s} b_{izs}^{0\text{–}5} \ge \frac{2}{3} \text{Pop}_z^{0\text{–}5}.$$

## (E) Minimum Distance Constraints

- Between New Facilities:

$$\sum_s y_{izs} + \sum_s y_{jzs} \le 1, \quad \forall z, (i,j) \in \mathcal{P}_z$$

- Between New and Existing Facilities:

$$\sum_s y_{izs} \le 0, \quad \forall z, (i,f) \in \mathcal{Q}_z$$

  (or simply remove $i$ from candidate list if $d_{if} < 0.06$ miles)

## (F)Variables (domains already $\ge 0$ unless noted)

- $x_f^{(k)} \ge 0$, $x_f = \sum_k x_f^{(k)}$, $x_f^{0\text{–}5} \ge 0$
- $y_{izs} \in \{0,1\}$
- $b_{izs}^{0\text{–}5} \ge 0$